A

PROJECT REPORT

ON

DESIGN OF FPV QUADROTOR FLIGHT CONTROLLER WITH DRONE TEST RIG

Submitted in partial fulfillment of the requirements for the award of degree of

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CERTIFICATE

This is to certify that the project titled **"Design of FPV Quadrotor Flight Controller with Drone Test Rig"** is a Bonafide work carried out, during the academic year **2025–2026**, under the supervision of **Dr. Niraj Kumar Goswami, Assistant Professor, School of Automation** from **Banasthali Vidyapith**.

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ABSTRACT

This project presents the design and development of a First Person View (FPV) quadrotor flight control system supported by a custom-built octagonal drone test rig. The objective is to construct a mini, lightweight, and affordable quadrotor with 5-inch propellers and an approximate weight of 1.5–2 kg, capable of providing stable flight performance. The quadrotor operates on the principle of thrust generation and vector control, wherein brushless DC motors and propellers create lift, while variations in motor speed regulate yaw, roll, and pitch. A Teensy4.0 flight controller is employed to ensure precise control and FPV capability.

To address the inherent risks of direct free-flight testing, a square first-flight tester gyro rig is integrated, providing three degrees of freedom for controlled evaluation of yaw, roll, and pitch responses. The square structure offers enhanced stability, uniform force distribution, and reliable motion capture compared to conventional test stands, thereby minimizing chances of damage and casualties. This methodology ensures safer validation of flight stability, enabling refinement of the control system before full deployment. The proposed system demonstrates a structured approach toward safe, reliable, and cost-effective quadrotor testing.

This abstract is hereby approved for submission.

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CHAPTER I

INTRODUCTION

1.1 BACKGROUND OF UAVS

A "drone" is an aerial vehicle without a pilot on board (an unmanned aerial vehicle). A drone is capable of flying a designated flight path as programmed in software or being controlled remotely. Global Positioning Systems (GPS) form the basis for tracking and controlling how technologies move in the world today.

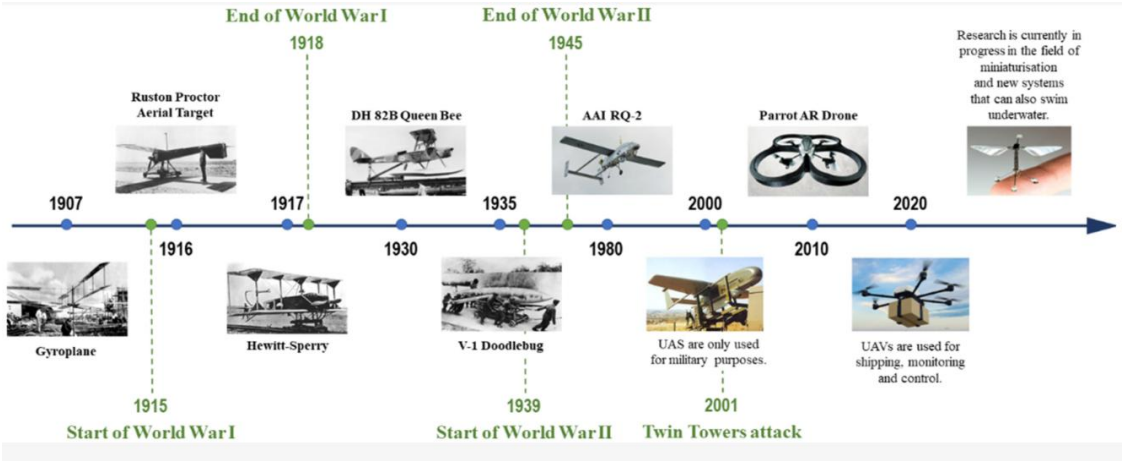

Fig [1] : Evolution of Drones

As shown in the timeline, the history of drones started in 1916 when the **Ruston Proctor Aerial Target** was invented and became the **first radio-controlled aircraft** and a stepping stone in developing modern drones. The other significant development related to unmanned flight was made about the same time, at the end of World War I, when Elmer Sperry, from the Sperry Gyroscope Company, introduced the **Hewitt-Sperry Automatic Aeroplan**. The plane acted as a basic prototype for the unmanned aerial drone, which flew autonomously, meaning it did not require a person to be physically involved in its navigation, thanks to gyroscopic stabilizers. At this early stage of drones' evolution, **their application was mainly military**; a number of prototypes were created, among which were different kinds of aircraft that could fly independently. Thus, the Standard E-1 planes were turned into autonomous monoplanes called the "**Larynx**" that could reach a pre-defined target. Post-World War I, innovations in the field kept going, and improvements were primarily made by the USA and Britain.

In 1935, the **DH 82B Queen Bee** was released as one of the first aircraft that could serve as **target drones for military training** and gave birth to the term "drone." Drone technology progressed

considerably around WWII, primarily because of the heightened demand from the military side. One of the examples of this development includes the **V-1 Doodlebug** – an early form of cruise missile that became the first aircraft capable of autonomous flight and was utilized as a flying bomb. There were many pioneers in the field of UAV creation; one of the best-known is probably Reginald Denny, who founded the company of "Reginald Denny Industries" and created **the first remote-controlled "Radioplanes"** for the U.S. military. Thousands of UAVs were built during the war years and modified for use in different scenarios for the military needs.

With the onset of the Cold War following World War II, the utilization of drones shifted from being employed as combat tools to being deployed for surveillance and reconnaissance missions, including radiation sampling missions. Drones had already been used widely as target drones for training purposes and to conduct intelligence missions without putting human pilots in danger. The late 1980s saw the development of advanced reconnaissance equipment such as the AAI RQ-2, thus improving the ability of drones to capture real-time intelligence. This was mainly done to monitor enemy movements.

Following the terrorist attacks on September 11th 2001, the use of drones increased greatly, marking **a new era of modern warfare**. Drones started being used not only for intelligence-gathering but also for combats due to their ability to conduct real-time surveillance and deliver precise attacks. The incorporation of GPS systems and video cameras enhanced the effectiveness of the missions conducted by drones. It should be noted that prior to 2000, the utilization of drones was strictly confined to military missions; however, this trend soon changed.

Starting from 2010, there has been a dramatic shift of drone technology towards its use for civilian and commercial purposes. It should be noted that the launch of the Parrot AR Drone in 2010 was the first step in the history of consumer-grade drones, as they were already operable through smartphones by means of Bluetooth technology. Over time, the use of drones became ubiquitous in many spheres of life, such as package delivery, surveillance, agricultural activities, mapping, and inspection of infrastructures. The development of drones is still ongoing, and the latest advancements include miniaturization, hybrid drones that can fly and dive, and the implementation of artificial intelligence to make the devices work autonomously.

Additionally, innovations in miniaturization will produce lightweight yet powerful drones, while hybrid systems will enable multi-environment operation, including air-to-water transitions. Improvements in battery technology and alternative energy sources such as hydrogen fuel cells will significantly increase flight endurance. Future drones will also play a key role in smart cities, being used for traffic monitoring, surveillance, logistics, precision agriculture, and infrastructure inspection. Overall, drone technology is evolving toward highly intelligent, energy-efficient, and multifunctional systems that will become an integral part of everyday life and advanced engineering applications.

1.2 CLASSIFICATION OF DRONES

The categorization of drones, also referred to as Unmanned Aerial Vehicles (UAVs), is crucial in determining the design and functionality of drones, besides their application areas. It should be noted that drones do not exist in one particular model or form. The drones are classified depending on various parameters like mode of flight, size, distance travelled, and application.

Categorizing drones will enable people to determine the best drone for a certain job depending on its use, ranging from taking aerial photographs, surveillance, farming, and even delivering items. Some of the broad classes of drones include multi-rotor drones, fixed wing drones, single rotor drones, among others.

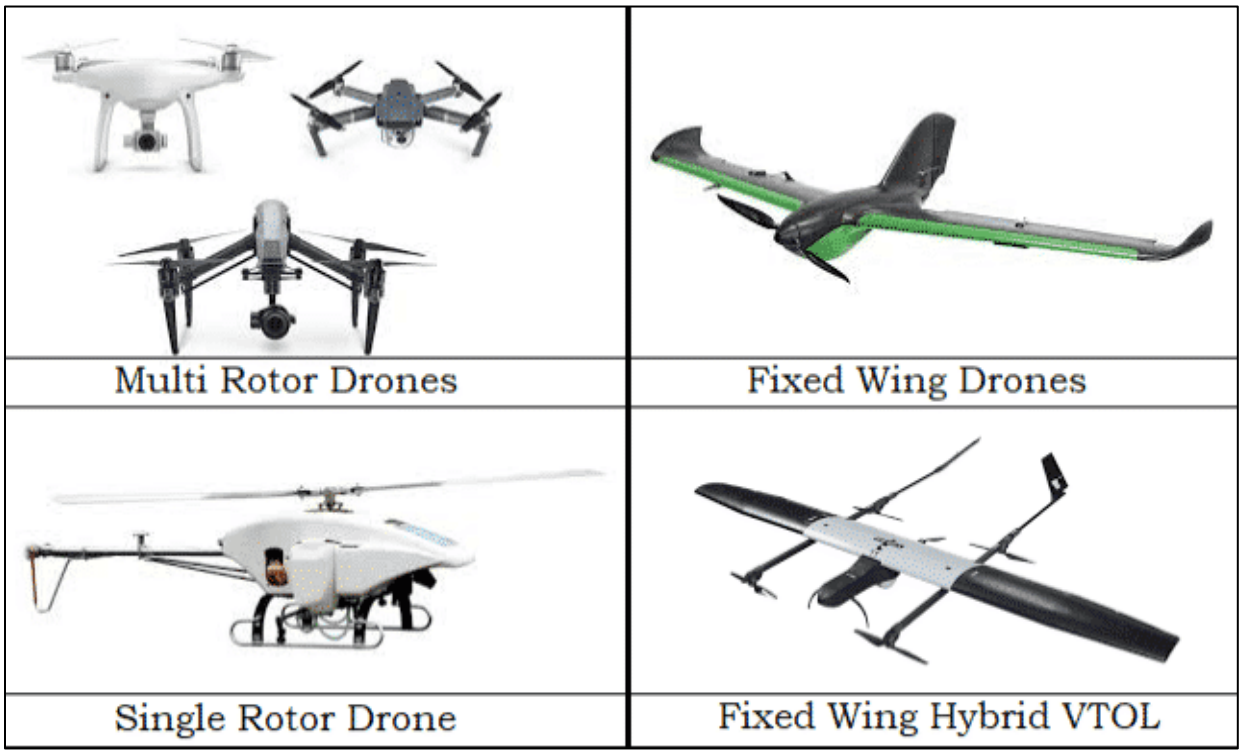

Fig [2]: Classification of Drones

Fixed Wing Drones:

Fixed-wing drones share similarities with regular airplanes. This means that these types of drones use aerodynamic lift from their wings to stay in the air and don't require hovering abilities. Compared to other drones which have rotor engines, fixed-wing models require forward movement and consume little energy. Therefore, they can fly for a longer time and over greater distances.

The main characteristics of fixed-wing drones are high flight endurance, increased speed of flights, and the ability to transport payloads over long distances. For these reasons, such drones are used for aerial mapping, surveying, agriculture, military operations, etc.

Multi-Rotor Drones:

Among UAV types, multi-rotor drones are the most common ones. These types of drones usually have four, six, or even eight propellers. The main advantage of multi-rotor drones is their vertical takeoff and landing (VTOL) abilities and stability. Multi-rotor drones can be easily controlled because they can stay in one place without any problems and hover there while using different speeds of their engines. In comparison with other drones, multi-rotor drones have low flight endurance because of high power consumption. Drones are widely employed in photography, reconnaissance, inspection, first-person view (FPV) racing, and even in micro-delivery services.

Single-Rotor Drones:

A single-rotor drone operates in the same way that helicopters do, relying on one large rotor blade for propulsion and hovering while a smaller tail rotor serves for stabilizing and controlling the direction of the aircraft. This design is more aerodynamically efficient compared to multi-rotors and is capable of lifting larger cargo using fewer resources. Nevertheless, this kind of drone is harder to handle due to its larger spinning blades. The key advantages include efficiency, endurance, and payload. These drones are typically used for various industrial purposes like heavy-lifting tasks, LiDAR surveys, extensive surveys, and military special operations.

Hybrid (VTOL) Drones:

The hybrid or VTOL drone combines the best of fixed-wing and multi-rotor aircraft. This means that such a drone is capable of vertical takeoff and landing like a multi-rotor drone and also of flying forward like a fixed-wing aircraft. Such design allows the drone to utilize the benefits of long-range flight of fixed-wing aircraft and hovering capabilities of multi-rotors. These characteristics include versatility, increased flight duration, and adaptability. The use of hybrid drones is becoming common in sophisticated missions like long-distance reconnaissance, cargo transport, remote area mapping, and military operations requiring both endurance and agility.

1.3 FPV (First-Person View) QUADROTOR DRONE

A first-person view quadrotor is a multi-rotor UAV that comes fitted with a real-time video transmission system to enable the operator to have an immersive and real-time experience of piloting the drone. "Quadrotor" describes the design of a multirotor UAV that involves fitting four motors in an X or + shape such that half of the rotors will be rotating in a clockwise direction while the others will be in counterclockwise rotation.

The FPV technology involved in this type of multi-rotor drone includes the use of a camera in the body of the drone, video transmitter (VTX), as well as a receiver with goggles or monitor. The basic concept involved in the operation of a quadrotor is altering the speed of individual motors to move in the direction of any of the three planes; namely, roll, pitch, and yaw. For instance, increasing motor speed at one side and reducing it at the other side results in the roll motion and vice versa.

FPV quadcopters are extensively employed because of their high maneuverability, small size, and capability of hovering with precision. They are utilized for FPV flying, filming with the help of cameras, observation tasks, structural inspections, and research purposes. Still, they possess some drawbacks, including short flight duration caused by high energy consumption and shorter range than fixed-wing aircraft. Nevertheless, their agility and controllability make them perfect for experiment purposes, mainly for those which require creation and examination of a custom flight controller.

Fig [3]: FPV Quadrotor

1.4 VTOL (Vertical Take-Off and Landing) DRONES

VTOL (Vertical Take-Off and Landing) drones represent sophisticated unmanned aerial vehicles that integrate advantages of both types – multi-rotor and fixed-wing aircraft. These UAVs are able not only to perform vertical take-off and landing just like a quadrotor but also to switch to forward flight mode similar to that of airplanes. Thus, VTOL drones help to solve the issues associated with traditional drones.

The main advantages of VTOL drones include greater endurance time, extended flight range, and maneuverability within confined spaces where there is no runway present. These types of drones are especially useful in applications like long-term monitoring, mapping remote locations, cargo delivery, disaster relief, and military missions. Moreover, VTOL drones are now being increasingly combined with emerging technologies like autonomous navigation systems and hybrid power units.

Despite having all these benefits, however, VTOLs are considerably more difficult to design and control than ordinary quadrotors. These drones need complicated control algorithms, more hardware components for transition processes, and improved aerodynamic properties. Nevertheless, despite all these difficulties, VTOL drones are definitely the drones of the future.

1.5 OBJECTIVE OF PROJECT

The main objective of this project is to build a customized FPV quadrotor flight controller that will be combined with an affordable drone testing facility in order to ensure a safe environment for testing purposes. In particular, the goal of the project is to create a stable and effective flight controller through real-time control algorithms and feedback sensors. Another important objective is to build a reliable test bench for experimenting with control parameters, including PID coefficients, without involving any risks that can occur during free flights. Moreover, this project is focused on developing knowledge related to embedded systems and real-time sensor processing in UAVs.

1.6 PROJECT SCOPE

The current project will be dedicated exclusively to the development and construction of a flight controller specifically designed for the operation of FPV quadcopters in a controlled setting. The scope of the project involves the design of the appropriate algorithm, the integration of required components, such as gyroscope and accelerometer sensors, and implementation of these elements via microcontrollers. Also included in the scope of the project will be the development of the appropriate drone test rig with restrictions of its translation motion, but free rotation. The project will involve only simple control of flights and will not encompass such elements as autonomous flight, obstacles avoidance, or swarm intelligence.

1.7 METHDOLOGY

The methodology used in this research project consists of a well-defined process that includes design, implementation, and testing phases. First, a comprehensive study into the physics of quadrotors and control theory concepts is done in order to identify the basic prerequisites of stabilization and controlling a flight vehicle. Once this information is acquired, hardware design is done, which involves choosing a proper microcontroller, installing sensors, and developing interfaces for communication purposes. The next step entails implementing a control system with the help of an embedded software solution and programming a flight controller through control algorithms such as PID to regulate roll, pitch, and yaw operations.

Once the flight control system is created, a mechanical test rig for drones is designed and built to ensure that the system can be tested safely since it limits the movement of the drone but allows rotation. The next step involves system integration and testing of the device to verify its operational capabilities. MATLAB Simulink and other similar software can be employed at the stage of testing the performance of the control strategy.

1.8 APPLICATION OF PROJECT

This UAV development project can find numerous practical uses and applications in both academia and research. The proposed solution can be applied as an educational tool for studying drone dynamics, control, and embedded systems programming. Due to the particular FPV quadrotor setup that is implemented in this UAV, it can be applied in different applications involving aerial photography, inspection, surveillance, and other operations that require high controllability. Moreover, the drone test rig designed during the process can be used in laboratory work when it comes to testing new strategies and approaches to UAV control that do not involve any flying operations. Finally, it can be the basis for future improvements and additions like UAV autonomous navigation, artificial intelligence incorporation, robotics, etc.

1.9 LIMITATION OF PROJECT

Although this project has some advantages, there are certain disadvantages that should be mentioned. First of all, the flight system was designed mainly for laboratory use and it might fail to reflect real flight conditions where there is turbulence and dynamic outdoor atmosphere. Moreover, there are no special algorithms and mechanisms for adaptive control or machine learning used in the flight controller design since it works with standard algorithms only, aiming at providing basic stabilization. Besides, the rig itself cannot perform translational movements, and thus, it prevents us from conducting a full analysis of flight behavior. It is important to mention that budget restrictions may influence the quality of high-end equipment chosen for the experiment.

CHAPTER II

DRONE COMPONENTS

2.1 INTRODUCTION

Drone (UAV), is a complex system that integrated mechanical, electrical and control algorithms to achieve controlled flight. The performance, stability, and efficiency of a drone depend heavily on the proper selection, design, and integration of its individual components.

This chapter focuses on the fundamental components that composes a multirotor drone, including the frame, propulsion system (motor and propellers), electronic subsystems like ESC (Electronic Speed Controller), light controller, power supply, power distribution systems and sensing and communication modules such as IMU(Inertial Measurement Unit), Radio transmitter- receiver system. Each of these components plays critical role in ensuring stable flight control and reliable operation under varying conditions.

These component's understanding not only help in assembling but through these, one can have insight into system level design considerations such as weight distribution, thrust generation, power management. Improper selection can lead to inefficiencies, instability, or can even cause whole system failure.

Therefore, this chapter aims to provide a comprehensive overview of drone's components, their working principles, key specifications, and their interdependencies within the overall system. This knowledge forms the foundation for designing, analyzing, and optimizing drone performance in practical applications.

The components of drone consists of :-

- Frame (Structure of Drone)
- Motors
- Propellers
- Electronics Speed Controllers
- Flight Controllers
- IMU
- Transmitter and Receiver
- Battery

2.2 FRAME (STRUCTURE OF DRONE)

S500 Quadcopter Fuselage Frame Kit PCB Version with Carbon Fiber Landing Gear Skid Kits.

This S500 Quadcopter frame is made from Glass Fiber which makes it tough and durable. They have the arms of ultra-durable Polyamide-Nylon which are the stronger molded arms having a very good thickness so no more arm breakage at the motor mounts on a hard landing.

The arms have support ridges on them, which improves stability and provides faster forward flight. It weighs around 405g only. The high-quality build offers best smooth flight unlike other low-quality frames.

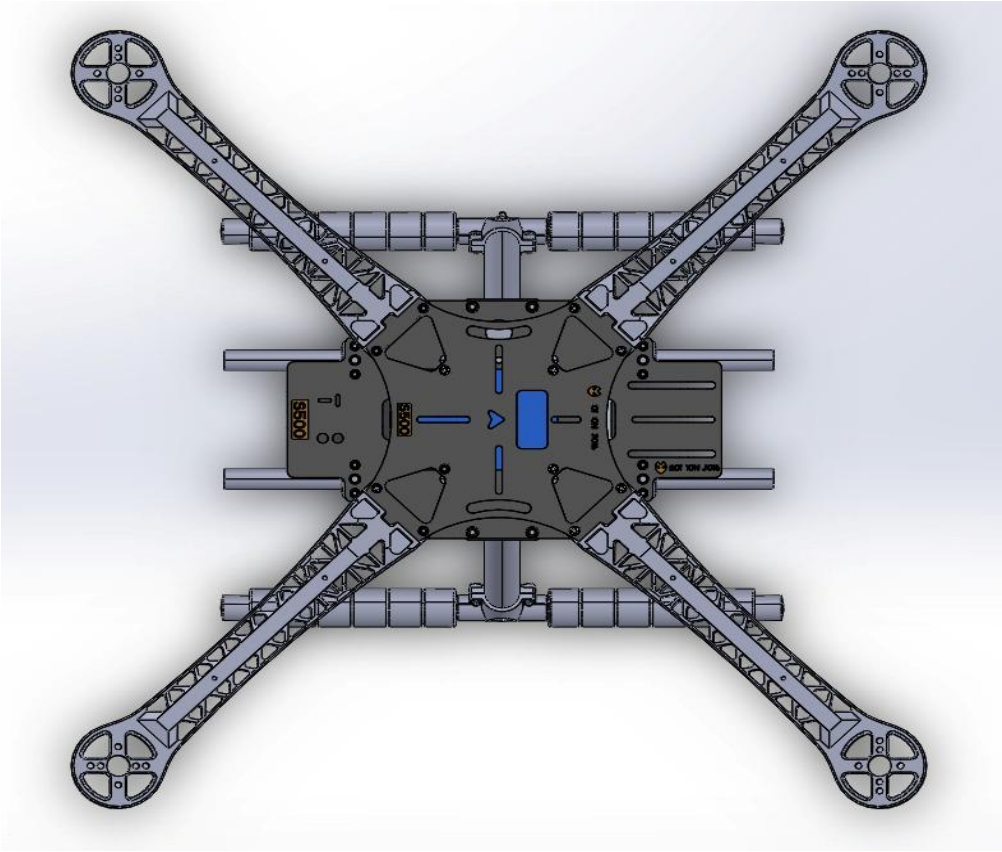

Fig 2.2.1 Cad of S500 quadrotor frame kit.

The S500 has a strong, light, and a sensible configuration including a PCB(Printed Circuit Board) with which you can directly solder your ESCs to the Quadcopter. So, making the Quadcopter build fast and easy.

So, it avoids the use of extra PDB(Power Distribution Board) and makes the mounting clean and neat. The S500 Quadcopter Frame is highly flexible frame during mounting of various components like flight controller, battery, etc.

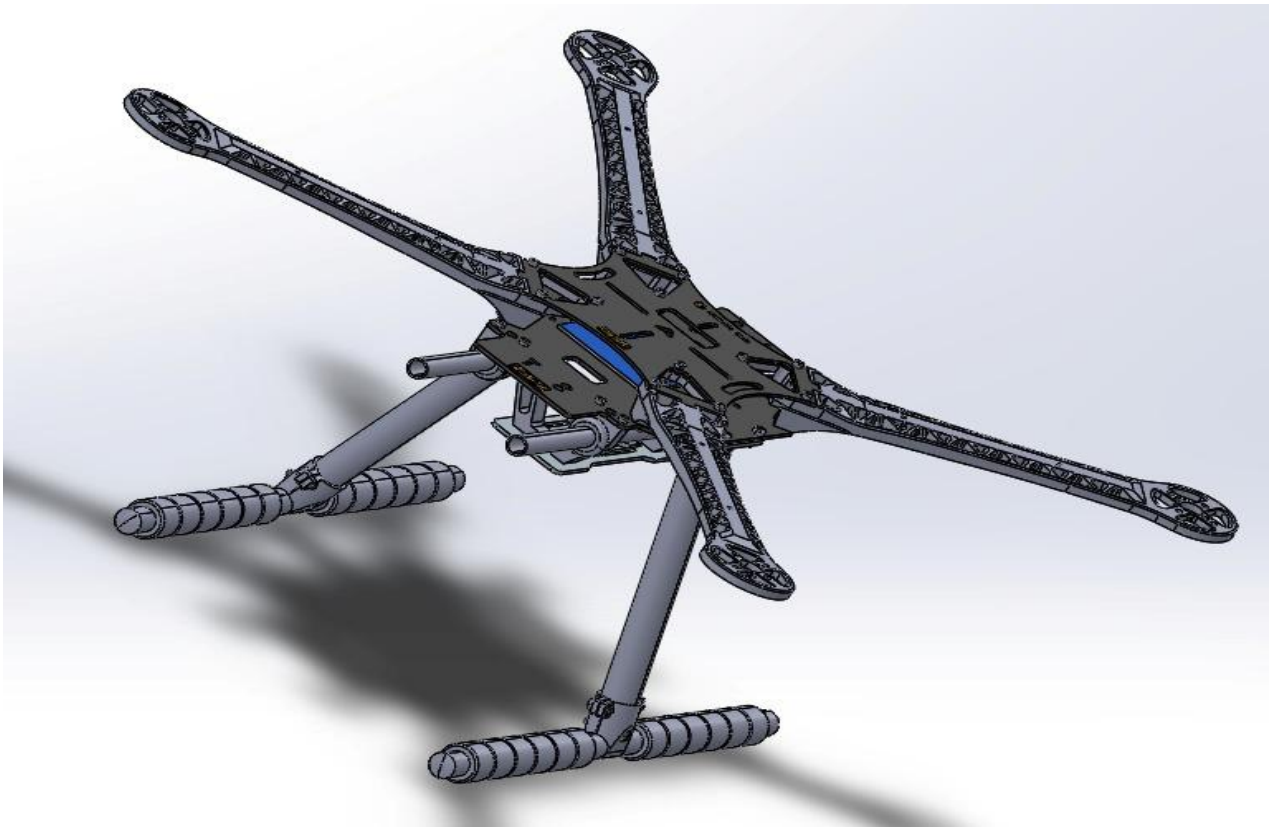

Fig 2.2.2 Cad of S500 quadrotor frame kit.

Features :

- Made by advanced engineering material, super strong & smooth
- Easy to assemble and disassemble.
- The frame is tough and durable.
- The main glass fiber frame has good strength.
- The arm equips support ridges for better and faster forward flight.
- Features pre-threaded brass sleeves.
- Large mounting tabs for camera mounting.
- In-built power distribution board.

2.3 MOTORS (BLDC Motor:- Brushless DC motor)

Quad-copter drone usually uses Brushless Direct Current (BLDC) motor as rotor due to high efficiency and small volume. BLDC motor is the motor does not have commutator and brushes. This reduces friction.

BLDC motor also is called synchronous motors, which use Direct Current electricity, by an inverter or switching power supply, which produces an Alternating Current to drive each phase.

Fig 2.3.1 BLDC Motor

The driver such ESC is used to control the speed also torque of the motor by the frequency and number of pulses of current to the motor windings.

The BLDC motor speed control is very important for drone position and velocity determent. The calculation of rotation speed, use of Pulse Width Modulation (PWM) technique for speed control and serial interface can be implemented using a microcontroller.

The following ratios for BLDC motor speed is similar to synchronous motor are:

$$N=60\ (f/P),$$

$$\omega = 2\pi\ (f/P),$$

where,

N= Motor speed in RPM;

ω=Motor speed in rad/s;

f = frequency of current pulses to winding in Hz;

P= number of pole pairs.

A quad copter is a multi-rotor drone, which is flied and propelled by four rotors or actuators. This type of drone usually applies two pairs of identical fixed pitched propellers, namely clockwise (CW) and counterclockwise (CCW).

The controlling of position can be achieved by variation of the speed of each rotor direction.

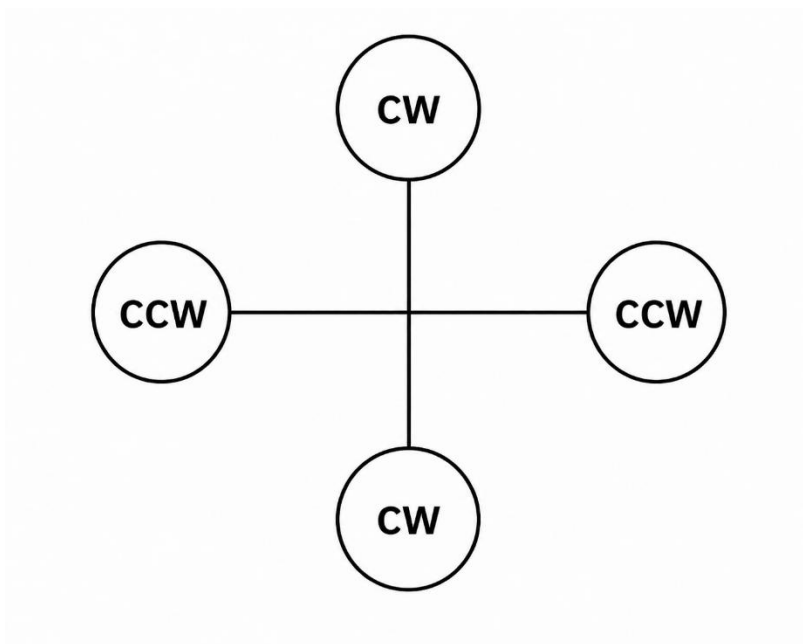

Fig 2.3.2 rotor direction of drone motor

Every rotor generates not only thrust but also torque around the center of rotation. These thrust creates a drag force oppositely to make the drone to fly.

Therefore, the speed control of each BLDC motor as quad copter drone rotor can control the flight position of drone accurately.

2.3.1 Operating dynamics of BLDC Motor in Drone:

The fundamental operation of BLDC motor is governed by Electromagnetism and the Lorentz Force.

- To rotate three-phase BLDC motors, six-step commutation is usually applied by allowing the current to flow in only two phases at every moment in time.
- The third phase is available to sense back EMF as rotor position indicator.
- The ESC driver is used to measure rotor position by hall sensor then it control the motor speed.
- Microcontroller sends the PWM pulses with adjusting duty cycle to transistors gates on ESC for motor speed control. The frequency of switching current flow through six-step commutation is proportional to the speed of motor.

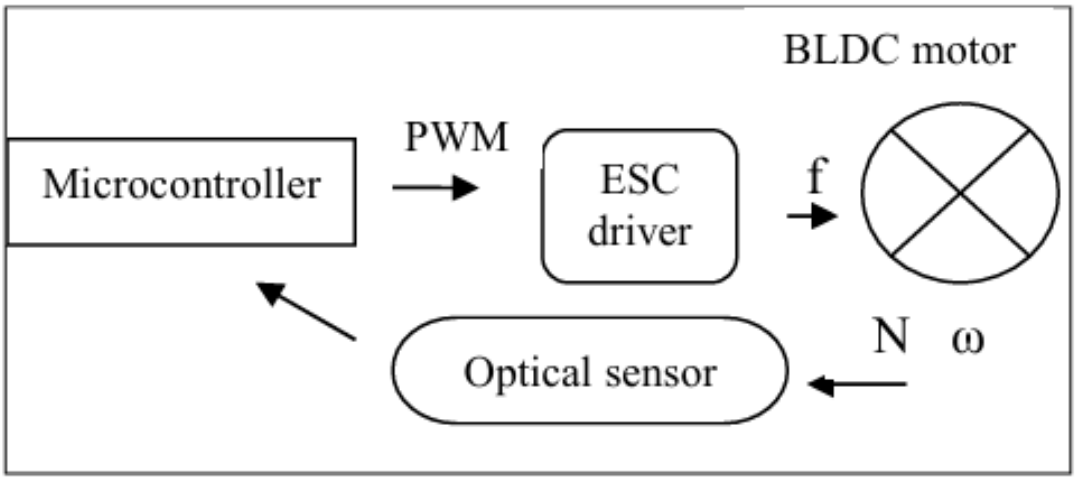

Fig 2.3.3 block diagram of instrumentation systems.

Features:

- High Power-to-Weight ratio.
- Absence of brushes eliminates friction and voltage drop.
- Rapid Speed Adjustments.
- High RPM Tolerance.
- Quick heat dissipate.
- Reduced EMI.

2.4 PROPELLER

Propellers are used to convert the rotational motion of BLDC motor into linear aerodynamic thrust. In a multirotor system, propellers are responsible for lift, directional control, and stabilization.

Fig 2.4.1 propeller

Propellers works on the Aerodynamic principles:

- Bernoulli's Principle: The curved upper surface of the propeller blade forces air to move faster over the top than the bottom. This creates lower pressure on top, resulting in upward lift (thrust).

- Angle of Attack (AoA): This is the angle between the blade's chord line and the oncoming air. An optimal AoA maximizes lift while minimizing the drag.

- Newton's Third Law: The propeller accelerates a mass of air downward; the reaction force pushes the drone upward.

2.4.1 Geometric Specifications

- Diameter : The distance from tip to tip. A larger diameter increases the "disc area," allowing the drone to move more air and hover more efficiently at the cost of responsiveness.

- Pitch: The theoretical distance a propeller would move forward in one full 360° rotation through a solid medium.

    1) Low Pitch: Provides higher "low-end" torque and better stability (standard for cinematic drones).
    2) High Pitch: Moves more air at high speeds, resulting in higher top speeds but lower efficiency at a hover (standard for racing drones).

- Blade Count:
    1) 2-Blade: Most efficient due to less "dirty air" (turbulence) from the preceding blade.
    2) 3-Blade: Provides more "grip" and smoother flight handling, though it draws more current.

2.4.2 Directional Logic and Yaw Control

Multirotor must cancel out the rotational torque generated by the motors to prevent the drone from spinning uncontrollably.

- CW vs. CCW: Drones use a combination of Clockwise (CW) and Counter-Clockwise (CCW) propellers.

- Torque Cancellation: In a quadcopter, two motors spin CW and two spin CCW. By balancing these forces, the drone remains straight.

- Yaw Maneuvers: To turn (yaw), the flight controller speeds up one diagonal pair of motors and slows down the other. This creates a torque imbalance that rotates the drone without changing its altitude.

2.4.3 Material science of propellers

Material	Advantages	Disadvantages
Plastic/Polycarbonate	Highly durable, cheap, bends during crashes.	Flexible, which can lead to "prop wash" or vibrations under high load.
Carbon Fiber	Extremely stiff, lightweight, high efficiency.	Expensive, brittle (shatters on impact), dangerous if it hits an object.
Glass Fiber Nylon	A middle ground for stiffness and cost.	Heavier than pure carbon fiber.

2.4.4 Performance Factors and Limitations

- Prop Wash: This occurs when a drone descends into its own turbulent air. The propellers lose lift because the air they are "grabbing" is already moving downward.
- Tip Vortices: High-pressure air leaks around the tip of the propeller to the low-pressure side, creating drag. Longer, thinner blades are generally more efficient at reducing these losses.
- Static vs. Dynamic Thrust: A propeller generates the most thrust when the drone is stationary (static). As the drone moves forward, the "effective" pitch changes, and thrust typically decreases.

2.5 ESC (Electronic Speed Controller)

The electric speed controllers (ESCs) take in the commands from the flight controller and adjust the speed of the individual motors accordingly. The ESCs control how fast the motors spin as well as increase and decrease the speed of specific motors to turn and bank in flight.

Fig 2.5.1 ESC

There are two main types of ESCs used for specific motor types:

- BRUSHED: The earlier design, brushed ESCs, are simpler and work with brushed DC motors. They control the speed of the motor through a process called pulse width modulation (PWM), which chops the voltage it receives from the battery. But because the brushes touch the motor, they wear out faster.

- BRUSHLESS brushless ESCs use electronic switches to generate a three-phase control AC current instead of physical brushes. This upgraded feature makes them more efficient, durable, and responsive, which is why they've become the standard for modern drones.

2.5.1 Operating Dynamics of ESC:

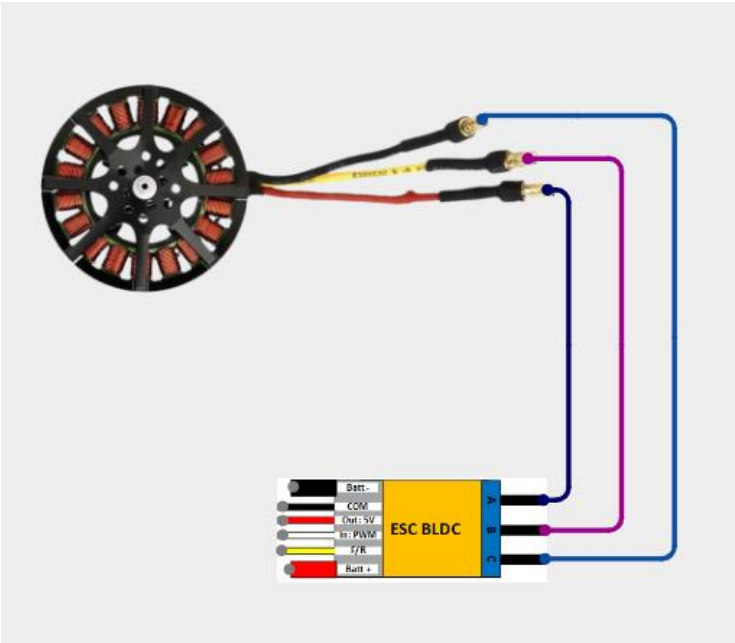

Fig 2.5.2 ESC with motor circuit

- First, the drone pilot inputs command signals using the flight controller.
- The drone then receives the control signal, such as a pulse-width modulation (PWM), a One-shot, or a MultiShot, through its onboard receiver.
- Next, the microcontroller interprets the signal and then adjusts the motor speed according to the intended output.
- The MOSFETs convert DC power from the battery into AC signals for brushless motors.
- The ESC adjusts voltage and current timing across the motor's three phases to control RPM. Higher frequencies lead to higher speeds and vice versa.
- ESCs use overcurrent protection, thermal shutdown, and low-voltage cutoff to safeguard both the motor and battery.

2.6 FLIGHT CONTROLLER

The Flight Controller (FC) is the "brain" of the drone. It is a small circuit board packed with sensors and a microprocessor that makes thousands of calculations per second to keep the aircraft stable and follow pilot commands.

The flight controller we used in drone is Teensy4.0. The Teensy 4.0 is a high-performance microcontroller board designed by PJRC and powered by the NXP i.MX RT1062 chip.

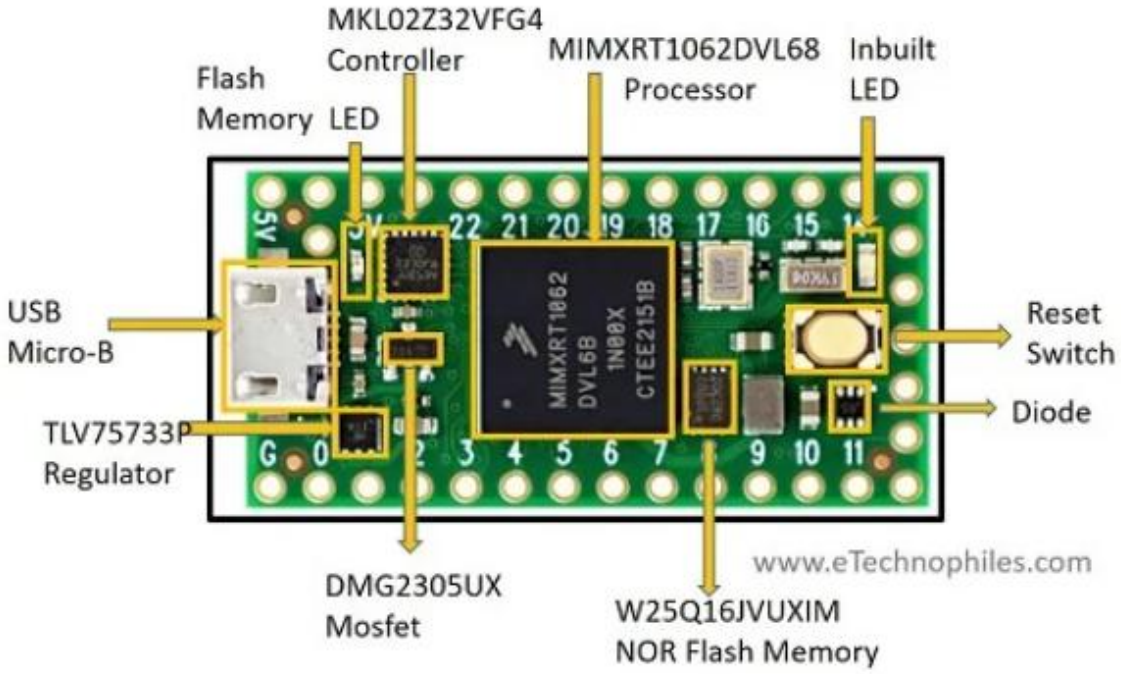

Fig 2.6.1 teensy board layout

- **Processor**: 600 MHz ARM Cortex-M7
- **RAM**: 1 MB (tightly coupled memory for high-speed access)
- **Flash Memory**: 2 MB
- **Operating Voltage**: 3.3V (I/O pins are not 5V tolerant)
- **USB Interface**: USB 2.0 (480 Mbps)
- **Digital I/O Pins**: 40 (with 31 supporting PWM)
- **Analog Input Pins**: 14 (12-bit ADC resolution)
- **Communication Protocols**: UART, SPI, I2C, CAN, I2S
- **Dimensions**: 1.4 x 0.7 inches (35.56 x 17.78 mm)

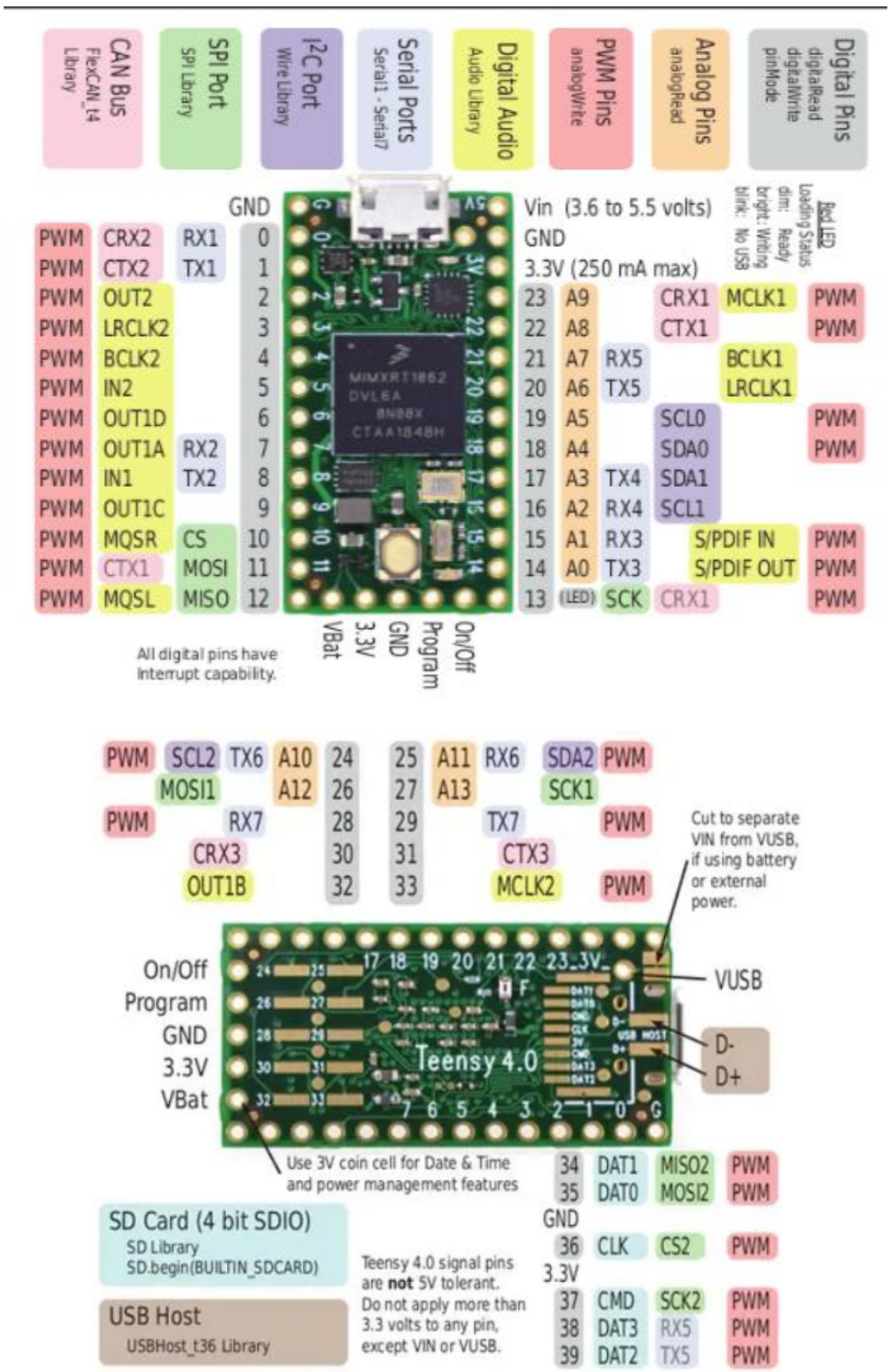

Fig 2.6.2 Pin diagram of teensy

Pin summary of teensy4.0

Function	Pins	Notes
Digital I/O	All 40 pins	3.3V output, not 5V tolerant
PWM Output	31 pins	19 independent frequency groups
Analog Input	14 pins	Up to 12-bit resolution (10-bit usable)
Serial (UART)	7 ports	All have FIFOs for high-speed performance
SPI	3 ports	First port has FIFO for faster transfers
I2C	3 ports	Supports 100, 400, and 1000 kbit/sec
CAN Bus	3 ports	1 port supports CAN-FD
I2S/TDM Audio	2 ports	Up to 16 channels per pin using TDM
S/PDIF	1 input, 1 output	Independent of I2S ports
USB Host	2 SMT pads (bottom)	No power management — do not hot-plug
LED	Pin 13	Orange LED, active HIGH

2.7 IMU (Inertial Measurement Unit)

An Inertial Measurement Unit (IMU) is a device that measures the angular velocity and linear acceleration in three axes of motion.

It is equipped with three-axis angular velocity (gyroscope) sensors, three-axis linear acceleration sensors, and a temperature sensor for measuring three-dimensional inertial motion (translational motion and rotational motion along three orthogonal axes). It measures linear acceleration and angular velocity with high accuracy to understand and enable control of attitude and trajectory for a moving body.

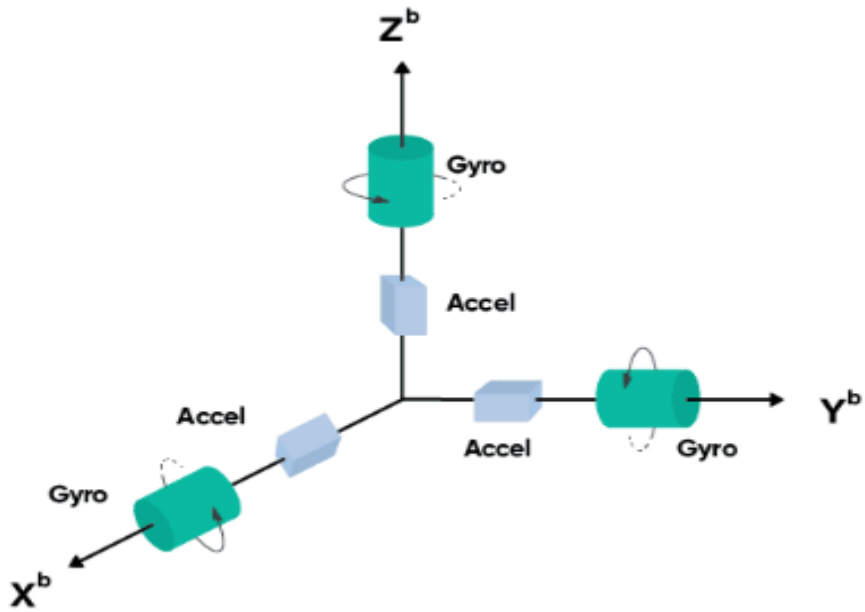

Fig 2.7.1 accelerometers and gyroscope of movement.

MPU9250 9-Axis Gyro Accelerator Magnetometer Module

MPU9250 is a multi-tasking sensor module based on MEMS (Micro Electro Mechanical Systems) architecture. It is a product by Asahi Kasei Microdevices Corporation. The MPU9250 is a 9-axial Motion Tracking device with a 3-Dimensional gyroscope, 3-Dimensional magnetometer, and 3-Dimensional accelerometer embedded into an ultra-compact IC.

The high-performance IC has 16-bit resolution analog-to-digital Converters. The sensitive module also has a VDDIO, an inbuilt temperature sensor, and an auxiliary I2C interface data transmission with non-inertial sensors.

The 9-DOF Sensor module comes with an I2C interface with a data rate of 400 kHz. A Serial Peripheral Interface for fast mode with a rate of up to 20 MHz is provided in the module. It also has frame synchronization.

MPU9250 Features and Specifications

Structure	MEMS
SPI Data Rate	1 Mhz
SPI Fast Data Reading Rate	20 Mhz
I2C Data Rate	400 kHz
Shock tolerance	10,000g
FIFO buffer	512 Bytes
Auxiliary I2C interface	1
SPI	1
I2C	1
Operating Voltage	2.4-3.6 Volts
Operating Current	3.5mA
VDDIO	Yes
Digital Output Temperature Sensor	Yes
Cross-Sensitivity	Minimum
User-Programmable Digital Filters	Yes

MPU9250 9-Axis Gyro Accelerator Magnetometer Module Pinout

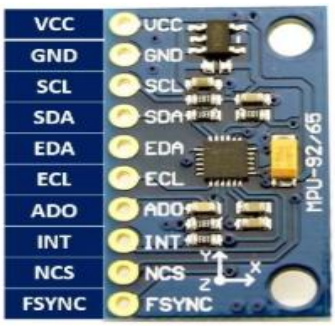

Fig 2.7.2 MPU pin configuration

Pin Name	Function
INT	Interrupt pin
ECL	I2C Master Serial Clock for external sensors connection pin
AD0/SDO	I2C Address/Serial data out pin
FSYNC	Frame Synchronization input pin
VCC	Power supply pin
GND	Ground pin
EDA	I2C Serial Data input for external sensors connection pin
nCS	Chip selection pin
SCL/SCLK	I2C Serial Clock/SPI Serial Clock pin
SDA/SDI	I2C Serial Data/SPI Serial Data pin

2.8 TRANSMITTER AND RECEIVER

In a drone system, communication between the pilot and the aircraft is done using a radio control (RC) system. The FlySky FS-CT6B is a commonly used RC transmitter for beginners and DIY drone builders.

This system consists of two main parts:

- Transmitter (TX) → Held by the pilot
- Receiver (RX) → Mounted on the drone

2.8.1 Receiver

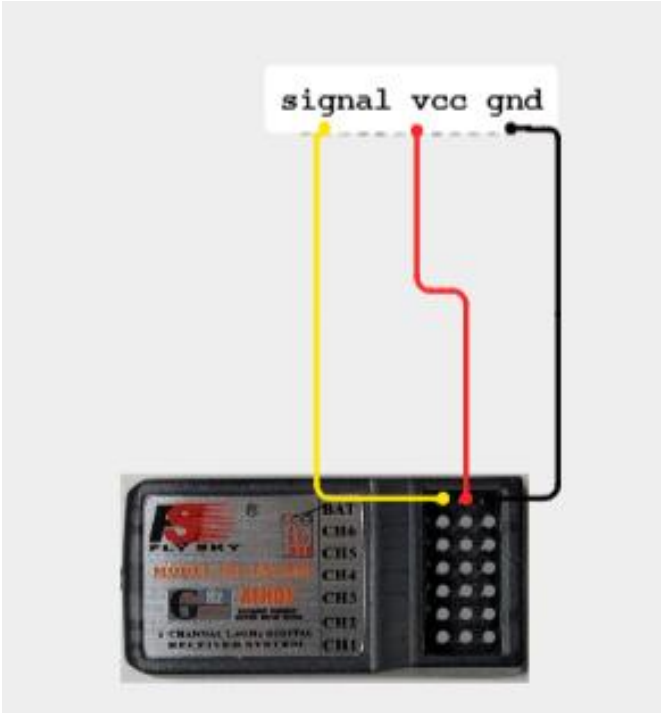

Fig 2.8.1 fly-sky receiver

The receiver is mounted on the drone and acts as an interface between the transmitter and the flight controller. It receives radio signals and converts them into electrical signals.

The key roles include:

- Receiving transmitted signals
- Decoding the signal information
- Outputting PWM or PPM signals to the controller

Channels Mapping

The FlySky FS-CT6B supports up to six channels, each assigned to a specific control function:

Channel	Function
CH1	Roll
CH2	Pitch
CH3	Throttle
CH4	Yaw
CH5	Auxiliary
CH6	Auxiliary

The receiver outputs PWM signals that are read by the flight controller, such as Teensy 4.0. Each channel corresponds to a specific control input.
Typical interfacing methods include:

- PWM signal reading using interrupts
- Pulse width measurement using microcontroller timers

The flight controller processes these inputs and generates appropriate control signals for the motors.

2.8.2 Transmitter

The transmitter is a fundamental component of a drone communication system, responsible for sending control commands from the pilot to the unmanned aerial vehicle (UAV). This section discusses the working principle, architecture, and role of the **FlySky FS-CT6B** transmitter. It explains how user inputs are converted into radio signals and transmitted to the receiver for real-time drone control.

The transmitter is a handheld device that converts physical joystick movements into wireless signals. It operates on a 2.4 GHz frequency band and transmits encoded signals corresponding to user commands.

The primary functions include:

- Generating control signals
- Encoding joystick inputs
- Wireless transmission to the receiver

Fig 2.8.2 fly-sky transmitter

Main Components:

- Control sticks (joysticks)
- Microcontroller (signal processing unit)
- RF transmission module
- Power supply (battery)
- Trimming and auxiliary switches

Working Principle

The operation of the transmitter involves multiple steps:

- The pilot moves the control sticks
- The internal microcontroller reads analog inputs
- Signals are converted into digital control data
- Data is encoded into a radio signal

The RF module transmits the signal wirelessly

CHAPTER III

DRONE TEST RIG

3.1 INTRODUCTION

A drone test rig is an essential experimental setup used to evaluate the performance and stability of a drone in a safe and controlled environment. Direct testing of drones in free flight can be risky, as it may lead to damage, unstable behavior, or safety hazards during initial trials and controller tuning.

To overcome these challenges, the drone test rig developed in this project is designed to restrict translational motion while allowing controlled rotational movement about multiple axes. This enables the study of important flight parameters such as roll, pitch, and yaw without the need for open-air testing.

The test rig is based on a multi-frame gimbal structure, where nested frames are connected through pivot joints to simulate real flight motion. When the drone generates thrust, the forces are transferred to the frame system, producing angular displacement instead of free movement. This allows observation and analysis of the drone's dynamic response, making it easier to evaluate stability and tune control systems such as PID controllers. The rigid outer frame ensures structural strength, while the inner frames provide flexibility for multi-axis motion.

Overall, the drone test rig serves as a reliable and cost-effective platform for testing, debugging, and improving drone performance. It bridges the gap between simulation and real-world implementation by providing a safe environment for repeated experimentation and system validation.

In addition to safety and control, the drone test rig also improves testing efficiency by allowing repeated experiments under identical conditions, which is difficult to achieve in free-flight scenarios.

3.2 TYPES OF DRONE TEST RIG

Drone test rigs are classified based on the number of degrees of freedom (DOF) they provide and the specific type of testing they are designed to perform. The degree of freedom refers to the number of independent directions or axes along which the drone is allowed to move or rotate during testing.

This classification is important because different test rigs are developed to study different aspects of drone performance, such as thrust generation, stability, control response, and dynamic behavior. Simpler rigs with fewer degrees of freedom are generally used for basic testing, such as evaluating motor performance or analyzing motion along a single axis, while more advanced rigs with multiple degrees of freedom are capable of simulating real flight conditions more accurately.

Additionally, the selection of a suitable test rig depends on factors such as the complexity of the experiment, required accuracy, safety considerations, and the level of control system analysis involved. Therefore, drone test rigs can range from simple fixed structures to complex multi-axis gimbal systems, each serving a specific purpose in the testing and development process. The commonly used types of drone test rigs are described below.:

- Single-Axis Test Rig (1-DOF)
- Two-Axis Test Rig (2-DOF)
- Three-Axis Test Rig (3-DOF)

3.2.1 Single-Axis Test Rig (1-DOF)

A Single-Axis Test Rig is the simplest form of drone testing setup in which the drone is allowed to rotate about only one axis, typically roll, pitch, or yaw. In this type of rig, the drone is mounted on a fixed structure with a pivot mechanism that permits motion in a single direction while restricting all other movements. This constrained setup helps in isolating and analyzing the behavior of the drone along one specific axis, making it easier to study fundamental performance characteristics.

Fig 3.2.1 single-axis (1-DOF) test rig.

Such rigs are commonly used for basic testing purposes, including thrust verification, motor performance analysis, and preliminary stability checks. Due to their simple design and minimal mechanical complexity, single-axis rigs are easy to fabricate, cost-effective, and suitable for initial stages of drone development. They also provide a safe testing environment by preventing uncontrolled motion during operation.

However, the major limitation of a single-axis test rig is that it cannot simulate the complete flight dynamics of a drone, as real drone motion involves simultaneous movement along multiple axes. Therefore, while it is useful for basic analysis, it is not suitable for advanced control system testing or full flight behavior evaluation.

3.2.2 Two-Axis Test Rig (2-DOF)

A Two-Axis Test Rig is designed to allow rotational motion about two independent axes, typically pitch and roll. In this configuration, the drone is mounted on a frame system that uses two perpendicular pivot joints, enabling movement in two directions while restricting motion along the third axis. This arrangement provides a better representation of drone behavior compared to a single-axis rig.

Fig 3.2.2 Two-Axis Test Rig (2-DOF)

Such rigs are widely used for analyzing balance, stability, and control response of the drone under partially realistic conditions. They are particularly useful for testing and tuning control algorithms, as they allow observation of how the drone reacts to disturbances in two axes simultaneously. The design is moderately complex, requiring careful alignment of joints to ensure smooth and accurate motion.

3.2.3 Three-Axis Test Rig (3-DOF)

A Three-Axis Test Rig is an advanced testing setup that allows rotational motion about all three axes—roll, pitch, and yaw—thereby closely simulating real flight conditions of a drone. In this type of rig, the drone is mounted within a multi-frame gimbal structure consisting of nested frames connected through pivot joints. Each frame is responsible for motion about a specific axis, enabling independent as well as combined rotational movements.

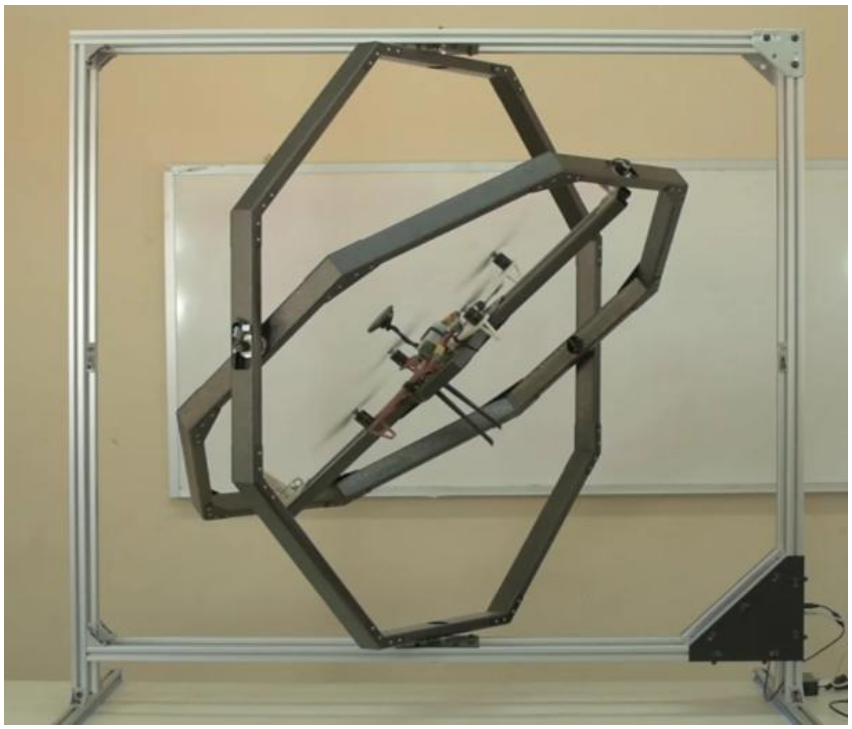

Fig 3.2.3 Eureka dynamics 3 dof drone test rig

This configuration provides a comprehensive platform for analyzing the dynamic behavior of the drone, including stability, control response, and interaction between different axes. It is especially useful for tuning advanced control systems such as PID controllers, as it allows realistic observation of how the drone responds to varying inputs and disturbances. The multi-axis capability makes it highly suitable for research, development, and performance validation.

However, the design and fabrication of a three-axis test rig are more complex compared to single- and two-axis rigs, requiring precise alignment and strong structural support. Despite this, it is considered the most effective type of test rig for detailed drone analysis, as it offers a close approximation of actual flight behavior while maintaining safety and control during testing.

3.3 JUSTIFICATION OF SELECTED DESIGN

The drone test rig developed in this project is based on a three-axis (3-DOF) gimbal structure, which was selected after evaluating different types of test rigs in terms of functionality, accuracy, and safety.

Compared to single-axis and two-axis rigs, the three-axis configuration provides a more comprehensive platform for analyzing drone behavior, as it allows rotational motion about all three axes—roll, pitch, and yaw.

This capability is essential for understanding real flight dynamics, where multiple motions occur simultaneously.

- It allows motion in roll, pitch, and yaw axes.
- It provides realistic simulation of drone behavior
- It is suitable for control system tuning and analysis
- It ensures safe and controlled testing conditions

The selected design enables detailed observation of the drone's stability and control response under constrained conditions. It is particularly suitable for tuning control algorithms such as PID controllers, as it allows realistic testing of how the drone reacts to input variations and external disturbances.

Additionally, the nested frame (gimbal) structure ensures smooth and independent motion along each axis, improving the accuracy of analysis. Another important reason for choosing this design is safety. The rig restricts translational movement, preventing uncontrolled flight and reducing the risk of damage during testing.

At the same time, it allows sufficient freedom of rotation to simulate actual flight conditions. The structure is also robust and stable due to the use of a rigid outer frame, which minimizes vibrations and enhances reliability.

Although the three-axis test rig is more complex to design and fabricate, its advantages in terms of realistic simulation, multi-axis analysis, and safe testing make it the most suitable choice for this project. Therefore, it effectively fulfills the requirements of drone performance evaluation and control system development.

3.4 DESIGN AND DIMENSIONS OF RIG

The drone test rig is designed as a **multi-axis gimbal structure** consisting of an outer supporting frame and a series of nested inner frames. The purpose of this design is to provide a rigid and stable structure while allowing controlled rotational motion about multiple axes. The outer frame acts as the main support and holds all internal components in position. Inside the outer frame, two square gimbals and one inner rectangular frame are mounted concentrically using pivot joints. Each frame is aligned along a different axis, enabling the simulation of roll, pitch, and yaw motions of the drone.

3.4.1 Outer Frame

The outer frame is the primary structural component of the drone test rig, providing overall support, stability, and rigidity to the entire system. It acts as the base structure that holds all inner gimbals in position and ensures minimal vibration during operation. The frame is designed to be strong enough to withstand forces generated by the drone while maintaining proper alignment of all components.

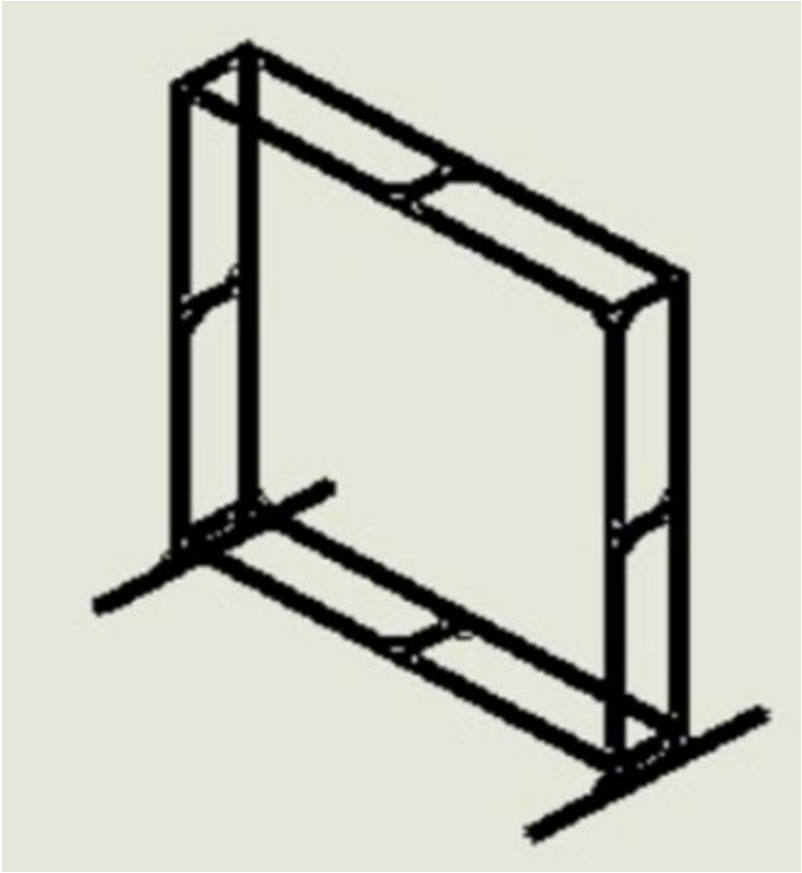

Fig 3.4.1 Cad model of outer frame

Dimensions

- **Outer Dimensions:** 121.5 × 121.5 cm²
- **Inner Dimensions:** 100.5 × 100.5 cm²

Key Details

- **Material Used:** Aluminium
- **Function:** Supports and encloses all inner frames
- **Features:** High strength, rigid structure, vibration resistance
- **Role:** Ensures stability and safe operation of the test rig

3.4.2 Outer Square Gimbal

The outer square gimbal is the first movable frame mounted inside the outer frame and plays a crucial role in enabling rotational motion of the test rig. It is connected to the outer frame through pivot joints, allowing rotation about one axis. This frame acts as an intermediate link between the fixed outer structure and the inner gimbals, ensuring smooth transmission of motion.

Dimensions

- **Outer Dimensions:** 90.5 × 90.5cm²
- **Inner Dimensions:** 82.5 × 82.5cm²

Key Details

- **Material Used:** Aluminium
- **Function:** Provides first level of rotational movement
- **Role:** Enables controlled motion and supports inner gimbals

3.4.3 Middle Square Gimbal

The middle square gimbal is mounted inside the outer square gimbal and provides the second level of rotational motion in the test rig. It is connected through pivot joints arranged perpendicular to the outer gimbal, allowing rotation about another axis. This frame plays a key role in enabling multi-axis movement and improving the realism of drone motion simulation.

Dimensions

- **Outer Dimensions:** 74.2 × 74.2 5cm²
- **Inner Dimensions:** 66 × 66cm²

Key Details

- **Material Used:** Aluminium

- **Function:** Provides secondary rotational motion.
- **Role:** Supports the inner frame.

3.4.4 Inner Rectangular Gimbal

The inner rectangular gimbal is the innermost frame of the test rig and is used to securely mount the drone. It is connected to the middle square gimbal through pivot joints, allowing rotation about the third axis. This frame directly interacts with the drone and is responsible for transmitting the effects of thrust and control inputs to the overall structure.

Dimensions

- **Outer Dimensions:** 55.2 × 28.1 5cm²
- **Inner Dimensions:** 47 × 20.01cm²

Key Details

- **Material Used:** Aluminium
- **Function:** Holds the drone and allows fine angular motion
- **Role:** Enables final axis movement and reflects drone behavior.
- **Features:** Compact design, secure mounting, smooth rotation

3.4.5 Central Mounting Platform

The central mounting platform is used to securely attach the drone to the inner rectangular gimbal and ensure proper alignment during testing. In this project, the platform is **3D printed using PTGF (Glass Fiber Reinforced Polymer)**, which provides a good balance of strength, rigidity, and lightweight characteristics. The platform consists of flat plates with provision for fastening, allowing the drone to be fixed using bolts. The cylindrical extension acts as a connector between the platform and the gimbal, enabling effective transfer of forces generated by the drone. The use of PTGF material improves durability and reduces overall weight, making the platform suitable for precise and reliable drone testing.

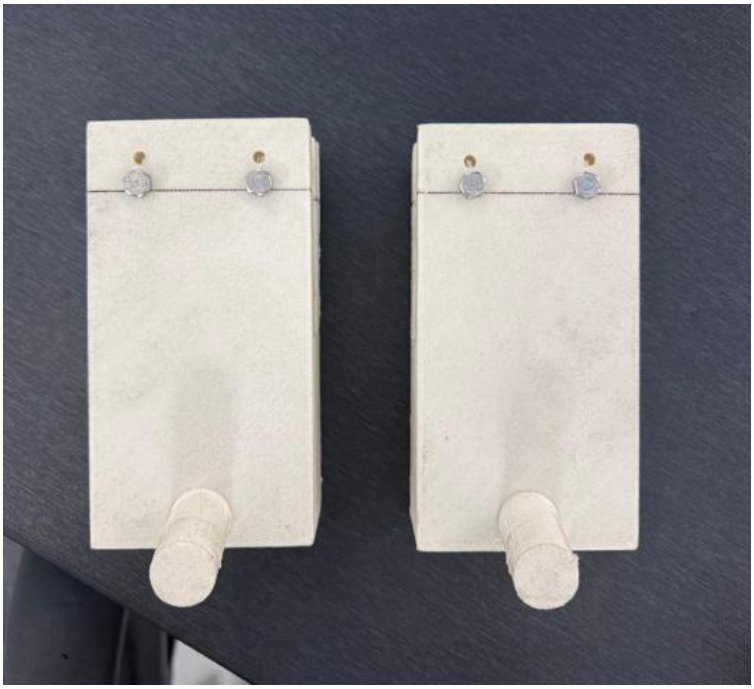

Fig 3.4.5 Central Mounting Platform

- **Material Used:** PTGF (Glass Fiber Reinforced 3D printed material)
- **Manufacturing Method:** 3D Printing
- **Structure:** Flat plate with fastening holes and cylindrical connector
- **Mounting Method:** Bolts and nuts
- **Function:** Securely holds the drone
- **Features:** Lightweight, high strength, good vibration resistance
- **Role:** Transfers thrust and motion to the gimbal system

3.4.6 Developed Drone Test Rig

The *Fig 5.4.5* above shows the final fabricated drone test rig developed in this project. All components, including the outer frame, gimbal structures, and central mounting platform, are assembled to form a complete and functional system. The rig is designed to provide a stable and controlled environment for testing drone performance.

Fig 3.4.5 final fabricated drone test rig developed in this project

- The assembled structure ensures proper alignment and integration of all components.
- The rig is compact and suitable for laboratory testing conditions.
- The design allows smooth operation and easy handling during experiments.
- The setup is robust and capable of withstanding forces generated during testing.
- It provides a practical platform for repeated and controlled experimentation.

3.5 WORKING PRINCIPLE

The working of the drone test rig can be understood with reference to the figure shown above. The drone is mounted on the **central mounting platform**, which is attached to the **inner rectangular gimbal**. This frame is connected to the **middle square gimbal**, which is further connected to the **outer square gimbal**, all supported by the rigid outer frame.

When the drone is powered ON, the propellers generate thrust, producing forces and moments. Since the outer frame restricts translational motion, these forces are transmitted to the gimbal system, resulting in rotational motion instead of free flight.

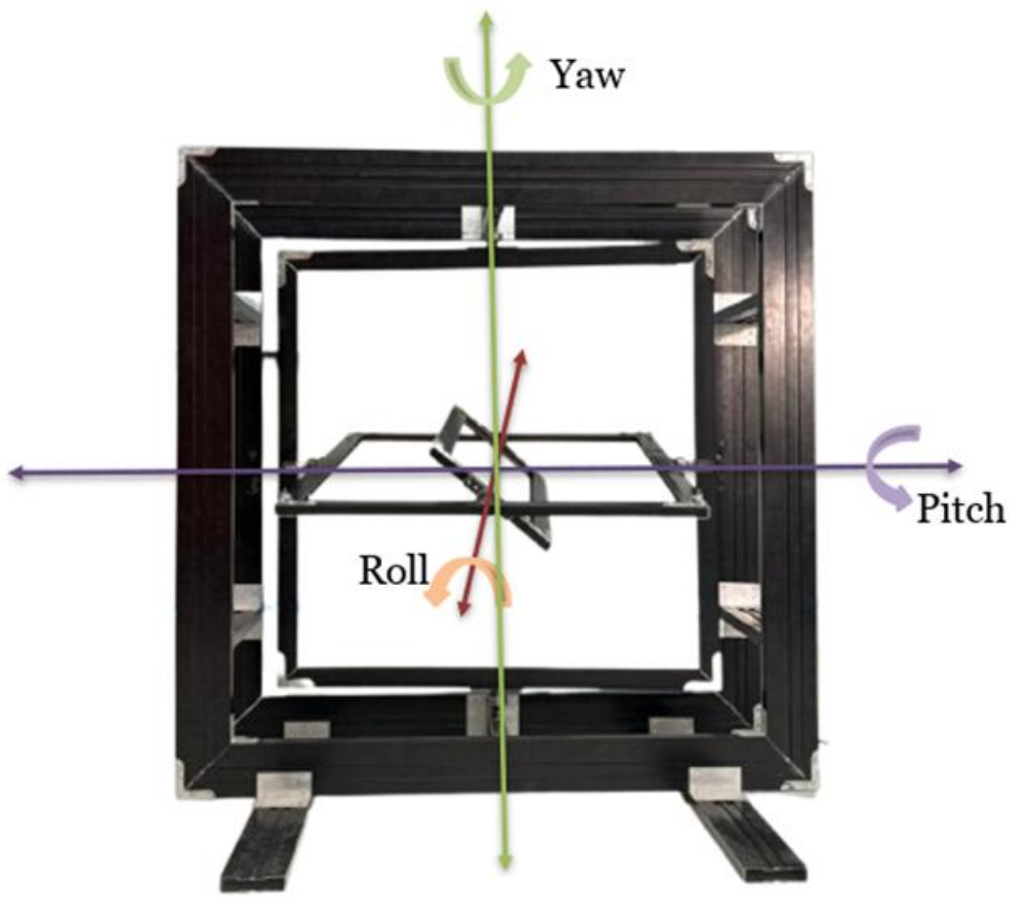

Fig 3.5 diagram of test rig

As observed in the figure:

- The outer square gimbal provides yaw motion (rotation about vertical axis)

- The middle square gimbal provides pitch motion (forward and backward tilt)
- The inner rectangular gimbal provides roll motion (side-to-side tilt)

Thus, each frame rotates about its respective axis, and the combined motion simulates real drone flight behavior. The movement of the frames reflects the drone's response to thrust and control inputs, allowing analysis of stability and performance in a safe and controlled environment.

Overall, the test rig converts the thrust generated by the drone into controlled multi-axis rotational motion.

3.6 ADVANTAGES

The drone test rig offers several benefits in terms of safety, performance, and usability:

- Provides a safe testing environment by restricting free flight
- Allows multi-axis motion (roll, pitch, yaw) for realistic simulation
- Reduces risk of drone damage and accidents
- Enables accurate observation of control response and stability
- Suitable for repeated and controlled experimentation
- Cost-effective compared to outdoor testing setups
- Compact and suitable for indoor laboratory use

3.7 APPLICATIONS

The drone test rig developed in this project can be used in various fields for testing and analysis purposes:

- Used for drone stability and performance testing in a controlled environment
- Applied in research and development of UAV systems
- Enables safe prototype validation without risk of damage
- Assists in testing sensors and flight controllers before actual flight
- Used for fault detection and troubleshooting in drone systems
- Supports aerodynamic behavior analysis in constrained conditions
- Can be used for vibration analysis and structural testing

The structured design helps in isolating specific parameters such as thrust variation, motor response, and controller performance, leading to more accurate analysis. Moreover, the compact and modular nature of the rig makes it suitable for laboratory use, where space and safety are important considerations. This makes the test rig a practical and valuable tool for both academic research and prototype development.

CHAPTER IV

FUTURE SCOPE

A drone test rig is an essential platform used to safely evaluate propulsion systems, flight controllers, sensors, and overall system stability before actual flight. As drone applications continue to expand across industries, the importance and sophistication of test rigs will increase significantly. The future scope of drone test rigs can be understood through the following developments:

1. Advanced Autonomous Testing

In the future, drone test rigs will become highly intelligent systems by integrating Artificial Intelligence (AI) and Machine Learning (ML) algorithms. These technologies will enable:

- Automatic fault detection without human intervention
- Real-time performance monitoring such as vibration, thrust, and efficiency
- Self-calibration, where the system adjusts itself for accurate testing

This will reduce human effort and increase testing accuracy and reliability.

2. Digital Twin and Simulation Integration

One of the most powerful advancements will be the use of digital twin technology. A digital twin is a virtual model of the physical drone system.

- The physical test rig will work alongside its digital twin
- Engineers can simulate failures before they happen in real life
- Helps in predictive maintenance and failure analysis

This approach will significantly reduce:

- Testing cost
- Development time
- Risk during real-world deployment

3. High-Precision Sensor Testing

Future drones rely heavily on advanced sensors, so test rigs will evolve to support precise sensor validation, including:

- LiDAR (Light Detection and Ranging) for mapping
- Computer vision modules for object detection
- Navigation systems that work without GPS

This is especially important for:

- Defense applications
- Autonomous delivery systems
- Indoor navigation drones

4. Multi-Drone System (Swarm) Testing

Drone technology is moving toward swarm systems, where multiple drones operate together.

Future test rigs will:

- Simulate multi-drone environments
- Test communication protocols between drones
- Evaluate coordination and collision avoidance

Applications include:

- Disaster management (search & rescue)
- Military operations
- Large-scale surveillance

5. Expansion in Industry Applications

Drone test rigs will not be limited to small UAVs; they will play a key role in larger and more complex systems:

- Aerospace certification testing
- Validation of urban air mobility systems (air taxis)
- Optimization of agricultural drones for spraying and monitoring

This will make test rigs critical tools in both research and industry.

6. Safety and Standardization

As drone usage increases, safety becomes more important. Future test rigs will contribute to:

- Development of standardized testing procedures
- Compliance with aviation authorities like
- DGCA (India)

This ensures:

- Safer drone operations
- Reliable commercial deployment
- Reduced chances of failure during flight

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